

Introduction to Statistical Methods
S2-25- AIMLZC418– Assignment 1

1. Each question carries 2.5 Marks ($2.5 \times 4 = 10$ Marks)
 2. Submissions are individual
 3. Solve on paper, scan and upload
 4. Plagiarism will result in zero marks
 5. Write your Name and BITS ID on each page
 6. Only handwritten solutions with complete steps, formulas and justification are accepted
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Q1. A startup recorded the daily number of app downloads for 15 days:

52, 48, 50, 49, 51, 53, 47, 52, 50, 49, 51, 48, 50, 300, 46

- a) Compute the mean, median and standard deviation.
- b) Detect whether an outlier exists using the IQR method.
- c) Remove the outlier (if any) and recompute the mean. Compare both means and comment on skewness.
- d) Which measure (mean/median) would you recommend for reporting performance to investors? Justify mathematically.

Q2: A cybersecurity system monitors network traffic and detects two types of threats: Malware (M) and Phishing (P).

Given:

$$\begin{aligned}P(M) &= 0.30 \\P(P) &= 0.25 \\P(M \cap P) &= 0.10\end{aligned}$$

Additionally:

- If a Malware attack occurs, the system correctly detects it with probability 0.85.
 - If a Phishing attack occurs, the system correctly detects it with probability 0.80.
 - If both attacks occur together, the system detects at least one of them with probability 0.95.
- (a) Find the probability that at least one type of threat occurs.
 - (b) Find the probability that exactly one threat occurs.
 - (c) Are Malware and Phishing independent events? Justify mathematically.
 - (d) Find the probability that a Malware attack occurs but is not detected.
 - (e) Find the probability that a Phishing attack occurs but is not detected.

- (f) Find the total probability that a threat occurs and is detected correctly.
- (g) Compute the overall probability that a threat occurs but the system fails to detect it.

Q3. A manufacturing company sources microchips from four suppliers:

Supplier	Supply %	Defective %
A	30%	2%
B	25%	3%
C	20%	5%
D	25%	4%

- (a) Find the probability that a randomly selected chip is defective.
- (b) If a chip is found defective, what is the probability it came from Supplier C?
- (c) If a chip is non-defective, find the probability it came from Supplier A.
- (d) Which supplier contributes most to defective chips? Justify using posterior probabilities.

Q4. A biased coin is tossed. The probability of getting Head is $P(H) = 0.7$ and $P(T) = 0.3$. After the coin is tossed, one card is drawn at random from a standard 52-card deck containing 26 Red cards and 26 Black cards.

The person observing the card reports its colour according to the following rules:

- If Head occurs:
The person reports the true colour of the card with probability 0.9 and reports the opposite colour with probability 0.1.
- If Tail occurs:
The person reports the opposite colour of the card with probability 0.8 and reports the true colour with probability 0.2.

Suppose the reported colour is “Red.”

Answer the following:

- a) Find the probability that the actual card drawn was Red.
- b) Find the probability that the coin showed Head given that “Red” was reported.
- c) Based on your answers, comment on whether the reporting system is reliable.

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